# MAE 472: Aerospace Structures II

## Final Project - May 4, 2026

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### Nomenclature

| | | |
|---|---|---|
| $n_{ply}$ | = | Number of Plies |
| $t_{tot}$ | = | Total Thickness of Beam (mm) |
| $t_{ply}$ | = | Thickness of Individual Ply (mm) |
| [Q] | = | Reduced Stiffness Matrix |
| $[\bar{Q}]$ | = | Transformed Stiffness Matrix |
| [A] | = | Extensional Stiffness Matrix (MPa * mm) |
| [B] | = | Coupling Stiffness Matrix (MPa * $mm^2$) |
| [D] | = | Bending Stiffness Matrix (MPa * $mm^3$) |
| {M} | = | Trial Bending Moment |
| $E_1$ | = | Longitudinal Elastic Modulus (MPa) |
| $E_2$ | = | Transverse Elastic Modulus (MPa) |
| $E_f$ | = | Flexural Modulus (MPa) |
| $G_{12}$ | = | In-Plane Shear Modulus (MPa) |
| $\nu_{12}$ | = | Major Poisson's Ratio |
| $\nu_{21}$ | = | Minor Poisson's Ratio |
| $S_L^+$ | = | Longitudinal Tensile Strength (MPa) |
| $S_L^-$ | = | Longitudinal Compressive Strength (MPa) |
| $S_T^+$ | = | Transverse Tensile Strength (MPa) |
| $S_T^-$ | = | Transverse Compressive Strength (MPa) |
| $S_{LT}$ | = | Shear Strength in Material System (MPa) |
| z | = | Distance from Laminate Midplane (mm) |
| k | = | Curvature |
| $\varepsilon_{xy}$ | = | Normal Strain in Global Coordinate System |
| $\gamma_{xy}$ | = | Shear Strain in Global Coordinate System |
| $\varepsilon_{12}$ | = | Normal Strain in Local Coordinate System |
| $\varepsilon^0$ | = | Midplane Strain |
| $\gamma_{12}$ | = | Shear Strain in Local Coordinate System |
| $\sigma$ | = | Normal Stress in Local Coordinate System (MPa) |
| $\tau$ | = | Shear Stress in Local Coordinate System (MPa) |
| $\theta$ | = | Ply Orientation Angle |
| $\alpha$ | = | Variable Ply Angle |
| m | = | Shorthand for $\cos(\theta)$ |
| n | = | Shorthand for $\sin(\theta)$ |
| $N_x$ | = | Uniaxial Force per Unit Length |

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# Problem 1

## 1.1 Methods

Composite laminates are widely used in the aerospace industry due to their high strength-to-weight ratios and directional stiffness characteristics. In this problem, Classical Laminate Theory is applied to determine the flexural modulus and the maximum allowable bending moment of carbon/epoxy laminate beams with $[0/90/0/90]_s$ and $[0/90/90/0]_s$ stacking sequences, and the two sequences are then compared. A uniaxial bending moment about the x-axis is assumed. Using the maximum stress criterion, we can also determine which ply fails first. The beam is 2 mm thick, 6 mm wide, and has 0.25 mm-thick plies. The flexural modulus is an intrinsic material property that determines how much a material can withstand bending under load. The maximum allowable bending moment is the amount of bending moment a material can withstand before exceeding allowable stress limits or causing permanent deformation.

### 1.1.1 Flexural Modulus

The beam has eight plies, determined by dividing the beam's total thickness by the thickness of an individual ply.

$$n_{ply} = t_{tot} / t_{ply}$$

**Equation I**

Then, the reduced stiffness matrix [Q] was found by using the material properties $E_1$ (180 GPa), $E_2$ (10 GPa), $G_{12}$ (7 GPa), and $\nu_{12}$ (0.28).

$$\nu_{21} = (E_2 / E_1) * \nu_{12}$$

$$Q_{11} = E_1 / (1 - \nu_{12} * \nu_{21})$$

$$Q_{22} = E_2 / (1 - \nu_{12} * \nu_{21})$$

$$Q_{12} = \nu_{12} * E_2 / (1 - \nu_{12} * \nu_{21})$$

$$Q_{66} = G_{12}$$

$$[Q] = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix}$$

**Equations II - VII**

Once [Q] is found, then the transformed stiffness matrix $[\bar{Q}]$ can be found for each ply based on the fiber orientations.

$$\bar{Q}_{11} = Q_{11}m^4 + 2(Q_{12} + 2Q_{66})m^2 n^2 + Q_{22} n^4$$

$$\bar{Q}_{22} = Q_{11} n^4 + 2(Q_{12} + 2Q_{66})m^2 n^2 + Q_{22} m^4$$

$$\bar{Q}_{12} = (Q_{11} + Q_{22} - 4Q_{66})m^2 n^2 + Q_{12}(m^4 + n^4)$$

$$\bar{Q}_{16} = (Q_{11} - Q_{12} - 2Q_{66})m^3 n - (Q_{22} - Q_{12} - 2Q_{66})m n^3$$

$$\bar{Q}_{26} = (Q_{11} - Q_{12} - 2Q_{66})m n^3 - (Q_{22} - Q_{12} - 2Q_{66})m^3 n$$

$$\bar{Q}_{66} = (Q_{11} + Q_{22} - 2Q_{12} - 2Q_{66})m^2 n^2 + Q_{66}(m^4 + n^4)$$

**Equations VIII - XIII**

Using $[\bar{Q}]$, [A], [B], and [D], the following are computed. [A], [B], and [D] represent extensional, coupling, and stiffness, respectively.

$$[A] = \Sigma (\bar{Q}_{nply} (z_{nply} - z_{nply-1}))$$

$$[B] = (1/2) \Sigma (\bar{Q}_{nply} ((z_{nply})^2 - (z_{nply-1})^2))$$

$$[D] = (1/3) \Sigma (\bar{Q}_{nply} ((z_{nply})^3 - (z_{nply-1})^3))$$

**Equations XIV - XVI**

The laminate is symmetric, which implies the coupling matrix, [B] = 0.

Finally, the flexural modulus can be found.

$$E_f = (12 * D_{11}) / t_{tot}^3$$

**Equation XVII**

### 1.1.2 Maximum Bending Stress

To determine the maximum allowable bending moment, a trial bending moment was applied, and the resulting curvature (k) was calculated. A trial bending moment is applied to the laminate to determine the stress distribution, enabling us to calculate the maximum bending moment using the linear relationship between the applied moment and the stress.

$$k = [D]^{-1}\{M\}$$

**Equation XVIII**

Then the strain at each ply was found as a function of its position through the thickness.

$$\varepsilon_x = z * \kappa_x$$

$$\varepsilon_y = z * \kappa_y$$

$$\gamma_{xy} = z * \kappa_{xy}$$

**Equations XIX - XXI**

After the global strains are identified, they can be transformed into local strains.

$$\varepsilon_1 = m^2\, \varepsilon_x + n^2\, \varepsilon_y + m\, n\, \gamma_{xy}$$

$$\varepsilon_2 = n^2\, \varepsilon_x + m^2\, \varepsilon_y - m\, n\, \gamma_{xy}$$

$$\gamma_{12} = -2\, m\, n\, \varepsilon_x + 2\, m\, n\, \varepsilon_y + (m^2 - n^2)\, \gamma_{xy}$$

**Equations XXII-XXIV**

The local stresses can then be found by multiplying the local strains by [Q].

$$\{\sigma_1, \sigma_2, \tau_{12}\} = [Q]\{\varepsilon_1, \varepsilon_2, \gamma_{12}\}$$

**Equation XXV**

Applying the maximum stress criterion, the stresses in each ply can determine failure to the corresponding material strength, and the applied bending moment was scaled linearly until the first ply reaches its failure limit. The five lamina strengths of this beam are: $S_L^+$ = 1800 MPa, $S_L^-$ = 1200 MPa, $S_T^+$ = 60 MPa, $S_T^-$ = 280 MPa, and $S_{LT}$ = 65 MPa.

$$- S_L^- < \sigma_1 < S_L^+$$

$$- S_T^- < \sigma_2 < S_T^+$$

$$|\tau_{12}| < S_{LT}$$

**Equations XXVI - XXVIII**

## 1.2 Numerical Results/Discussion

| Layup | Flexural Modulus (MPa) | Max Moment (N·mm) | First Failure | Failure Mode |
|---|---|---|---|---|
| [0/90/0/90]s | 127430 | 3865 | Top ply | $\sigma_1$ |
| [0/90/90/0]s | 111423 | 3379 | Top ply | $\sigma_1$ |

**Table 1**

After following the above procedure, the flexural modulus for the $[0/90/0/90]_s$ lamina is 127430.029 MPa, and the maximum bending moment is 3865.549 N-mm. The top ply fails first, and it fails the $\sigma_1$ criterion. After repeating the process for the $[0/90/90/0]_s$, the flexural modulus is 111422.808 MPa, and the maximum bending moment is 3379.792 N-mm. The top ply also fails in this lamina, and it again fails the $\sigma_1$ criterion. This shows that the stacking sequence significantly affects the beam's overall structural integrity. The $[0/90/0/90]_s$ lamina has a higher flexural modulus because the 0° plies are farther from the midpoint, increasing stiffness. The stiffness increases because it is proportional to the distance from the neutral axis; placing the stiffer plies farther from the midplane increases the moment of inertia, which in turn increases the rigidity. The lamina with the higher flexural modulus can withstand a greater bending moment, as expected. Failure was observed in the outer plies due to their greater distance from the midplane, which causes them to experience the highest stress. Also, the failure was attributed to longitudinal stress, indicating that fibers aligned with the loading direction govern the lamina's overall strength. This shows that laminate stacking sequences can be optimized to maximize stiffness and load capacity without increasing material usage, which is crucial in aerospace design.

# Problem 2

### 2.1.1 Comparing the Local and Global Stresses within each Ply of Two Laminas under a Uniaxial Load

In this problem, a symmetric angle-ply laminate is subjected to a single uniaxial force per unit length, $N_x$ = 80 MPa, with $N_y$ = 0 and $N_{xy}$ = 0. The thickness of each ply is 25 mm, and the lamina properties are: $E_1$ = 158 GPa, $E_2$ = 19 GPa, $G_{12}$ = 6.9 GPa, $\nu_{12}$ = 0.3. Using these material properties, we can calculate the resulting stresses in each ply at different angles. The process is the same from problem 1 to problem 2, up to and including the computation of the [A], [B], and [D] matrices. In problem one, we use [D] to compute the curvature. Still, when the load is in-plane, the laminate response is computed using the extensional matrix [A] rather than the curvature stiffness matrix [D].

$$\varepsilon^0 = [A]^{-1}\{N_x\}$$

**Equation XXIX**

We can then find the global stresses by multiplying the midplane strain by $[\bar{Q}]$.

$$\{\sigma_x, \sigma_y, \tau_{xy}\} = [\bar{Q}]\{\varepsilon^0\}$$

**Equation XXX**

The local stresses are determined using the same formulas as in problem 1. It is important to note that the strain used in this problem comes from the midplane strain, not the curvature.

### 2.1.2 Find the Global and Local Stresses under a Uniaxial Force as the Interior Ply Orientations Increase

Redo the process from problem 2.1.1 each time the interior ply orientation changes.

### 2.1.3 Plot

Plot $\sigma_x$ as a function of $\alpha$

### 2.2.1 Problem 2, Part a: Numerical Results/Discussion

The first lamina has an orientation of [+55/-55/-55/+55].

| Ply | Angle_deg | sigma_x | sigma_y | tau_xy | sigma_1 | sigma_2 | tau_12 |
|---|---|---|---|---|---|---|---|
| 1 | 55 | 80 | 0 | 9.452160107 | 35.20131937 | 44.79868063 | -40.82053399 |
| 2 | -55 | 80 | 0 | -9.452160107 | 35.20131937 | 44.79868063 | 40.82053399 |
| 3 | -55 | 80 | 0 | -9.452160107 | 35.20131937 | 44.79868063 | 40.82053399 |
| 4 | 55 | 80 | 0 | 9.452160107 | 35.20131937 | 44.79868063 | -40.82053399 |

**Table 2**

In this first lamina, the global stress is uniform, with $\sigma_x$ = 80 MPa, as expected since the uniaxial load was applied only in the x-direction. $\tau_{xy}$ has the same magnitude for all four plies, but opposite sign for -55 due to the fiber orientations. When expressed in the material coordinate system, the same trend holds. $\sigma_1$ and $\sigma_2$ have the same values for all four plies, and $\tau_{12}$ has the same magnitude for all four plies, but opposite sign for the ply orientation. This is consistent with classical laminate theory, where opposite fiber orientations produce opposite shear stresses.

### 2.2.2 Problem 2, Part b: Numerical Results/Discussion

The second lamina has an orientation of [+6/-55/-55/+6]

| Ply | Angle_deg | sigma_x | sigma_y | tau_xy | sigma_1 | sigma_2 | tau_12 |
|---|---|---|---|---|---|---|---|
| 1 | 6 | 139.6053575 | -4.455459739 | 10.21940962 | 140.156055 | -5.006157224 | -4.979873042 |
| 2 | -55 | 20.39464251 | 4.455459739 | -10.21940962 | 19.30239415 | 5.547708101 | 10.98421016 |
| 3 | -55 | 20.39464251 | 4.455459739 | -10.21940962 | 19.30239415 | 5.547708101 | 10.98421016 |
| 4 | 6 | 139.6053575 | -4.455459739 | 10.21940962 | 140.156055 | -5.006157224 | -4.979873042 |

**Table 3**

The second lamina has a different stress distribution due to the changes in the fiber orientation. The outer plies endure a much higher $\sigma_x$ compared to the inner plies since the outer plies are more closely aligned with the direction of the load. In contrast, the inner plies are more misaligned, causing the load to be transferred through shear. For the local stresses, the values of $\sigma_1$ and $\sigma_2$ change only slightly, but the same trend is prevalent. The outer plies endure most of the stress due to the small fiber orientation. The fibers are nearly aligned with the force, so they bear most of the stress. The inner plies experience a stronger $\tau_{12}$ because the inner plies are much more misaligned with the direction of the axial force, causing the load to be redistributed via shear.

### 2.2.3 Problem 2, Part c and d: Numerical Results/Discussion

This next lamina has an orientation of [6/-α/-α/6] where α increases by 3 degrees from 3 to 30.

| alpha_d | Pl | Angle_( | sigma_x | sigma_y | tau_xy | sigma_1 | sigma_2 | tau_12 |
|---|---|---|---|---|---|---|---|---|
| 3 | 1 | 6 | 76.7689512 | 0.26415954 | 5.49183421 | 77.0748611 | -0.04175037 | -2.58129584 |
| 3 | 2 | -3 | 83.2310488 | -0.26415954 | -5.49183421 | 83.5764041 | -0.60951479 | -1.09793646 |
| 3 | 3 | -3 | 83.2310488 | -0.26415954 | -5.49183421 | 83.5764041 | -0.60951479 | -1.09793646 |
| 3 | 4 | 6 | 76.7689512 | 0.26415954 | 5.49183421 | 77.0748611 | -0.04175037 | -2.58129584 |
| 6 | 1 | 6 | 80 | 4.4409E-16 | 7.45177405 | 80.675215 | -0.67521497 | -1.02753272 |
| 6 | 2 | -6 | 80 | 4.4409E-16 | -7.45177405 | 80.675215 | -0.67521497 | 1.02753272 |
| 6 | 3 | -6 | 80 | 4.4409E-16 | -7.45177405 | 80.675215 | -0.67521497 | 1.02753272 |
| 6 | 4 | 6 | 80 | 4.4409E-16 | 7.45177405 | 80.675215 | -0.67521497 | -1.02753272 |
| 9 | 1 | 6 | 84.5243811 | -0.40502728 | 9.20722262 | 85.5107146 | -1.39136083 | 0.17711427 |
| 9 | 2 | -9 | 75.4756189 | 0.40502728 | -9.20722262 | 76.483699 | -0.60305284 | 2.84245522 |
| 9 | 3 | -9 | 75.4756189 | 0.40502728 | -9.20722262 | 76.483699 | -0.60305284 | 2.84245522 |
| 9 | 4 | 6 | 84.5243811 | -0.40502728 | 9.20722262 | 85.5107146 | -1.39136083 | 0.17711427 |
| 12 | 1 | 6 | 89.7748699 | -0.90770579 | 10.6833521 | 91.0052478 | -2.13808367 | 1.02291144 |
| 12 | 2 | -12 | 70.2251301 | 0.90770579 | -10.6833521 | 71.5740378 | -0.4412019 | 4.33724042 |
| 12 | 3 | -12 | 70.2251301 | 0.90770579 | -10.6833521 | 71.5740378 | -0.4412019 | 4.33724042 |
| 12 | 4 | 6 | 89.7748699 | -0.90770579 | 10.6833521 | 91.0052478 | -2.13808367 | 1.02291144 |
| 15 | 1 | 6 | 95.3000464 | -1.46598943 | 11.8566387 | 96.7078952 | -2.8738382 | 1.53814764 |
| 15 | 2 | -15 | 64.6999536 | 1.46598943 | -11.8566387 | 66.3924005 | -0.22645752 | 5.54034072 |
| 15 | 3 | -15 | 64.6999536 | 1.46598943 | -11.8566387 | 66.3924005 | -0.22645752 | 5.54034072 |
| 15 | 4 | 6 | 95.3000464 | -1.46598943 | 11.8566387 | 96.7078952 | -2.8738382 | 1.53814764 |
| 18 | 1 | 6 | 100.783927 | -2.04432716 | 12.736893 | 102.308554 | -3.56895409 | 1.76896327 |
| 18 | 2 | -18 | 59.2160729 | 2.04432716 | -12.736893 | 61.2432148 | 0.01718518 | 6.49799156 |
| 18 | 3 | -18 | 59.2160729 | 2.04432716 | -12.736893 | 61.2432148 | 0.01718518 | 6.49799156 |
| 18 | 4 | 6 | 100.783927 | -2.04432716 | 12.736893 | 102.308554 | -3.56895409 | 1.76896327 |
| 21 | 1 | 6 | 106.02864 | -2.61497372 | 13.3512773 | 107.617465 | -4.20379854 | 1.76538113 |
| 21 | 2 | -21 | 53.9713603 | 2.61497372 | -13.3512773 | 56.3095317 | 0.27680229 | 7.26013244 |
| 21 | 3 | -21 | 53.9713603 | 2.61497372 | -13.3512773 | 56.3095317 | 0.27680229 | 7.26013244 |
| 21 | 4 | 6 | 106.02864 | -2.61497372 | 13.3512773 | 107.617465 | -4.20379854 | 1.76538113 |
| 24 | 1 | 6 | 110.924516 | -3.15711994 | 13.7332516 | 112.53334 | -4.76594478 | 1.57369423 |
| 24 | 2 | -24 | 49.0754845 | 3.15711994 | -13.7332516 | 51.6847886 | 0.54781581 | 7.87265855 |
| 24 | 3 | -24 | 49.0754845 | 3.15711994 | -13.7332516 | 51.6847886 | 0.54781581 | 7.87265855 |
| 24 | 4 | 6 | 110.924516 | -3.15711994 | 13.7332516 | 112.53334 | -4.76594478 | 1.57369423 |
| 27 | 1 | 6 | 115.421244 | -3.65523673 | 13.9161915 | 117.01353 | -5.24752224 | 1.23339314 |
| 27 | 2 | -27 | 44.5787559 | 3.65523673 | -13.9161915 | 47.4025523 | 0.83144033 | 8.37417906 |
| 27 | 3 | -27 | 44.5787559 | 3.65523673 | -13.9161915 | 47.4025523 | 0.83144033 | 8.37417906 |
| 27 | 4 | 6 | 115.421244 | -3.65523673 | 13.9161915 | 117.01353 | -5.24752224 | 1.23339314 |
| 30 | 1 | 6 | 119.505208 | -4.09739608 | 13.9303449 | 121.050983 | -5.64317091 | 0.77672022 |
| 30 | 2 | -30 | 40.494792 | 4.09739608 | -13.9303449 | 43.4594756 | 1.13271252 | 8.79536232 |
| 30 | 3 | -30 | 40.494792 | 4.09739608 | -13.9303449 | 43.4594756 | 1.13271252 | 8.79536232 |
| 30 | 4 | 6 | 119.505208 | -4.09739608 | 13.9303449 | 121.050983 | -5.64317091 | 0.77672022 |

**Table 4**

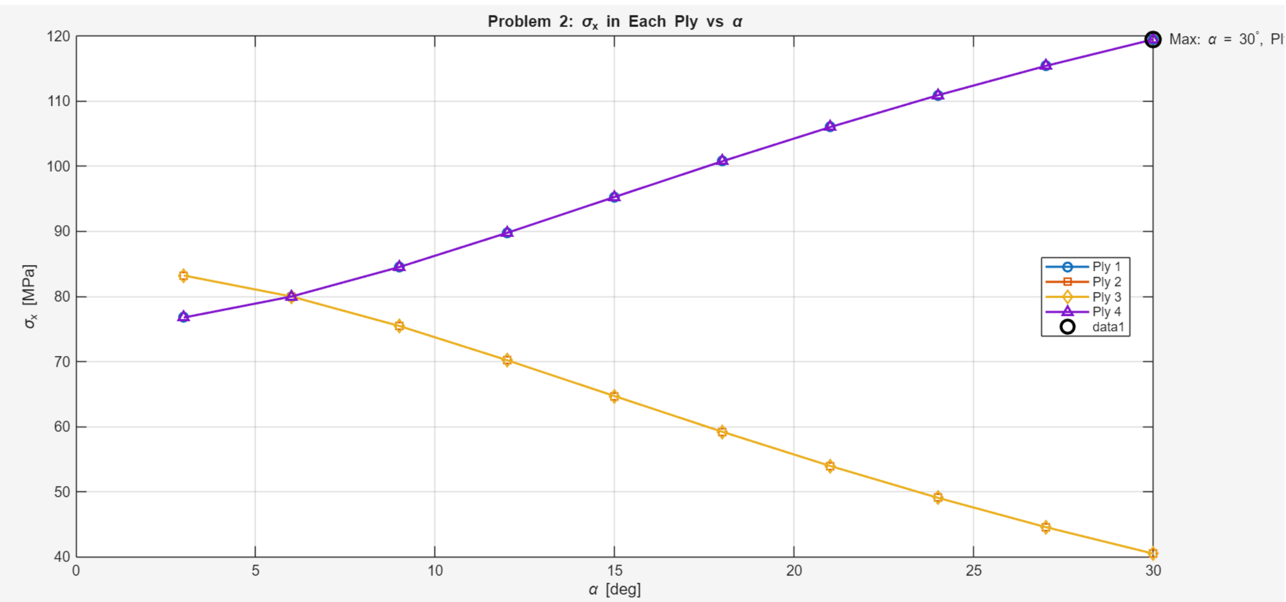


**Figure 1: Variation of $\sigma_x$ in each ply as a function of inner ply orientation angle α.**

The stresses in each ply were calculated for varying values of the inner ply orientations. The results are summarized in Table 3, and the variation of $\sigma_x$ in each ply as a function of $\alpha$ is shown in Figure 1. The stress distribution changes significantly as $\alpha$ changes. The inner plies endure a higher $\sigma_x$ until $\alpha$ reaches -6. If the inner plies have a smaller orientation, they will endure a higher longitudinal stress. After the inner ply orientation exceeds 6 degrees, the outer plies endure a greater longitudinal stress. This is expected in classical lamina theory. Most of the stress is in the longitudinal direction, since the uniaxial stress is also in that direction. The outer ply $\sigma_x$ begins at about 76 MPa and increases to about 11

MPa. In contrast, the inner plies show a decrease in $\sigma_x$, beginning at about 83 MPa and dropping to 40 MPa as their orientation increases. This is clearer in Figure 1, as both ply 1 and 4 have a near-linear increase in stress as $\alpha$ increases, and ply 2 and 3 have an opposite trend. This shows how the load is redistributed as the ply orientations change and how it is evenly distributed when the ply orientations are the same. Using this predictable relationship between fiber orientation and load distribution, we can optimize laminate designs. As the inner ply orientations become more misaligned, the outer plies experience greater stress because they are aligned with the loading direction, thereby increasing shear stress, as shown in Table 3. The maximum $\sigma_x$ is found to be when $\alpha$ is at its greatest, and in the outer plies. This behavior models anisotropy, in which increasing fiber misalignment reduces axial stiffness and shifts the load to more-aligned plies. This proves that increasing the angle in the inner plies decreases their load-bearing capacity, whereas it increases it in the outer plies. Overall, this demonstrates that ply orientation is detrimental to determining a laminate's load capacity. These results show the importance of laminate design in optimizing structural performance.

# 1. Appendix

### 3.1 Extra Stuff

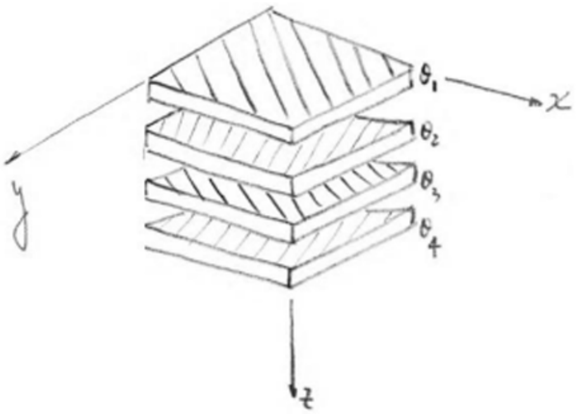


**Figure 2: Drawing of laminate for problem 2**

### 3.2 Code

MATLAB code used for all calculations is provided below.

```
%% Nathan McIntire 200485406
% 5/4/2026
% MAE 472: Final Project
clear; clc; close all
%% Problem 1 - Given values
XX = 6;                 % last 2 digits of student ID are 06
width = XX;             % mm
E1 = 180000;            % MPa
E2 = 10000;             % MPa
G12 = 7000;             % MPa
v12 = 0.28;
SL_tension = 1800;      % MPa
SL_compression = 1200;  % MPa
ST_tension = 60;        % MPa
```

```
ST_compression = 280;    % MPa
SLT_shear = 65;          % MPa
total_thickness = 2;     % mm
ply_thickness = 0.25;    % mm
number_of_plies = total_thickness / ply_thickness;
%% Layup for part a and b
theta = [0 90 0 90 90 0 90 0];
%% Create Q matrix
v21 = v12 * E2 / E1;
Q11 = E1 / (1 - v12*v21);
Q22 = E2 / (1 - v12*v21);
Q12 = v12*E2 / (1 - v12*v21);
Q66 = G12;
Q = [Q11 Q12 0;
   Q12 Q22 0;
   0   0   Q66];
%% z locations through the thickness
z = linspace(-total_thickness/2, total_thickness/2, number_of_plies+1);
%% Build A, B, and D matrices
A = zeros(3,3);
B = zeros(3,3);
D = zeros(3,3);
for k = 1:number_of_plies
  angle = theta(k);
  m = cosd(angle);
  n = sind(angle);
  Qbar11 = Q11*m^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*n^4;
  Qbar22 = Q11*n^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*m^4;
  Qbar12 = (Q11 + Q22 - 4*Q66)*m^2*n^2 + Q12*(m^4 + n^4);
  Qbar16 = (Q11 - Q12 - 2*Q66)*m^3*n - (Q22 - Q12 - 2*Q66)*m*n^3;
  Qbar26 = (Q11 - Q12 - 2*Q66)*m*n^3 - (Q22 - Q12 - 2*Q66)*m^3*n;
  Qbar66 = (Q11 + Q22 - 2*Q12 - 2*Q66)*m^2*n^2 + Q66*(m^4 + n^4);
  Qbar = [Qbar11 Qbar12 Qbar16;
       Qbar12 Qbar22 Qbar26;
       Qbar16 Qbar26 Qbar66];
  z_bottom = z(k);
  z_top = z(k+1);
  A = A + Qbar*(z_top - z_bottom);
  B = B + 0.5*Qbar*(z_top^2 - z_bottom^2);
  D = D + (1/3)*Qbar*(z_top^3 - z_bottom^3);
end
%% Flexural modulus
flexural_modulus = 12*D(1,1)/(total_thickness^3);
fprintf('Flexural modulus = %.3f MPa\n', flexural_modulus)
%% Maximum bending moment using maximum stress criterion
moment_per_width = 1;        % trial moment per unit width
moment_vector = [moment_per_width; 0; 0];
curvature = D \ moment_vector;
failure_ratio = zeros(number_of_plies,3);
for k = 1:number_of_plies
  angle = theta(k);
  z_middle = (z(k) + z(k+1))/2;
  strain_x = z_middle * curvature(1);
  strain_y = z_middle * curvature(2);
  gamma_xy = z_middle * curvature(3);
  m = cosd(angle);
  n = sind(angle);
  strain_1 = m^2*strain_x + n^2*strain_y + m*n*gamma_xy;
```

```
  strain_2 = n^2*strain_x + m^2*strain_y - m*n*gamma_xy;
  gamma_12 = -2*m*n*strain_x + 2*m*n*strain_y + (m^2 - n^2)*gamma_xy;
  stress_local = Q * [strain_1; strain_2; gamma_12];
  sigma_1 = stress_local(1);
  sigma_2 = stress_local(2);
  tau_12 = stress_local(3);
  if sigma_1 >= 0
    failure_ratio(k,1) = sigma_1 / SL_tension;
  else
    failure_ratio(k,1) = abs(sigma_1) / SL_compression;
  end
  if sigma_2 >= 0
    failure_ratio(k,2) = sigma_2 / ST_tension;
  else
    failure_ratio(k,2) = abs(sigma_2) / ST_compression;
  end
  failure_ratio(k,3) = abs(tau_12) / SLT_shear;
end
largest_ratio = max(failure_ratio,[],'all');
max_moment_per_width = moment_per_width / largest_ratio;
max_bending_moment = max_moment_per_width * width;
[row, col] = find(failure_ratio == largest_ratio);
first_failed_ply = row(1);
if col(1) == 1
  failure_type = 'sigma_1';
elseif col(1) == 2
  failure_type = 'sigma_2';
else
  failure_type = 'tau_12';
end
fprintf('\nMaximum bending moment = %.3f N-mm\n', max_bending_moment)
fprintf('First failed ply = %d\n', first_failed_ply)
fprintf('Failure type = %s\n', failure_type)
%% Problem 1 Part C
% New layup: [0/90/90/0]s
theta = [0 90 90 0 0 90 90 0];
%% Rebuild A, B, and D matrices for the new layup
A = zeros(3,3);
B = zeros(3,3);
D = zeros(3,3);
for k = 1:number_of_plies
  angle = theta(k);
  m = cosd(angle);
  n = sind(angle);
  Qbar11 = Q11*m^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*n^4;
  Qbar22 = Q11*n^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*m^4;
  Qbar12 = (Q11 + Q22 - 4*Q66)*m^2*n^2 + Q12*(m^4 + n^4);
  Qbar16 = (Q11 - Q12 - 2*Q66)*m^3*n - (Q22 - Q12 - 2*Q66)*m*n^3;
  Qbar26 = (Q11 - Q12 - 2*Q66)*m*n^3 - (Q22 - Q12 - 2*Q66)*m^3*n;
  Qbar66 = (Q11 + Q22 - 2*Q12 - 2*Q66)*m^2*n^2 + Q66*(m^4 + n^4);
  Qbar = [Qbar11 Qbar12 Qbar16;
          Qbar12 Qbar22 Qbar26;
          Qbar16 Qbar26 Qbar66];
  z_bottom = z(k);
  z_top = z(k+1);
  A = A + Qbar*(z_top - z_bottom);
  B = B + 0.5*Qbar*(z_top^2 - z_bottom^2);
  D = D + (1/3)*Qbar*(z_top^3 - z_bottom^3);
```

```
end
%% Flexural modulus for Part C
flexural_modulus_c = 12*D(1,1)/(total_thickness^3);
fprintf('\nProblem 1 Part C Results\n')
fprintf('Flexural modulus for [0/90/90/0]s = %.3f MPa\n', flexural_modulus_c)
%% Maximum bending moment for Part C
moment_per_width = 1;
moment_vector = [moment_per_width; 0; 0];
curvature = D \ moment_vector;
failure_ratio = zeros(number_of_plies,3);
for k = 1:number_of_plies
   angle = theta(k);
   z_middle = (z(k) + z(k+1))/2;
   strain_x = z_middle * curvature(1);
   strain_y = z_middle * curvature(2);
   gamma_xy = z_middle * curvature(3);
   m = cosd(angle);
   n = sind(angle);
   strain_1 = m^2*strain_x + n^2*strain_y + m*n*gamma_xy;
   strain_2 = n^2*strain_x + m^2*strain_y - m*n*gamma_xy;
   gamma_12 = -2*m*n*strain_x + 2*m*n*strain_y + (m^2 - n^2)*gamma_xy;
   stress_local = Q * [strain_1; strain_2; gamma_12];
   sigma_1 = stress_local(1);
   sigma_2 = stress_local(2);
   tau_12 = stress_local(3);
   if sigma_1 >= 0
       failure_ratio(k,1) = sigma_1 / SL_tension;
   else
       failure_ratio(k,1) = abs(sigma_1) / SL_compression;
   end
   if sigma_2 >= 0
       failure_ratio(k,2) = sigma_2 / ST_tension;
   else
       failure_ratio(k,2) = abs(sigma_2) / ST_compression;
   end
   failure_ratio(k,3) = abs(tau_12) / SLT_shear;
end
largest_ratio_c = max(failure_ratio,[],'all');
max_moment_per_width_c = moment_per_width / largest_ratio_c;
max_bending_moment_c = max_moment_per_width_c * width;
[row_c, col_c] = find(failure_ratio == largest_ratio_c);
first_failed_ply_c = row_c(1);
if col_c(1) == 1
   failure_type_c = 'sigma_1';
elseif col_c(1) == 2
   failure_type_c = 'sigma_2';
else
   failure_type_c = 'tau_12';
end
fprintf('Maximum bending moment for [0/90/90/0]s = %.3f N-mm\n', max_bending_moment_c)
fprintf('First failed ply = %d\n', first_failed_ply_c)
fprintf('Failure type = %s\n', failure_type_c)
%% Compare the two layups
fprintf('\nComparison of Problem 1 Layups\n')
fprintf('[0/90/0/90]s max bending moment = %.3f N-mm\n', max_bending_moment)
fprintf('[0/90/90/0]s max bending moment = %.3f N-mm\n', max_bending_moment_c)
if max_bending_moment > max_bending_moment_c
   fprintf('[0/90/0/90]s can withstand the larger bending moment.\n')
```

```
else
  fprintf('[0/90/90/0]s can withstand the larger bending moment.\n')
end
%% Problem 2
% Calculate stresses in each ply for different angle-ply laminates
clearvars -except XX
clc
fprintf('\n================ Problem 2 ================\n')
%% Given material properties for Problem 2
E1 = 158000;        % MPa
E2 = 19000;         % MPa
G12 = 6900;         % MPa
v12 = 0.30;
ply_thickness = 0.25; % mm
Nx = 80;            % MPa-mm
Ny = 0;
Nxy = 0;
force_vector = [Nx; Ny; Nxy];
%% Create Q matrix
v21 = v12 * E2 / E1;
Q11 = E1 / (1 - v12*v21);
Q22 = E2 / (1 - v12*v21);
Q12 = v12*E2 / (1 - v12*v21);
Q66 = G12;
Q = [Q11 Q12 0;
   Q12 Q22 0;
   0   0   Q66];
%% Problem 2 Part A
% Layup: [+55/-55/-55/+55]
theta = [55 -55 -55 55];
number_of_plies = length(theta);
total_thickness = number_of_plies * ply_thickness;
A = zeros(3,3);
for k = 1:number_of_plies
  angle = theta(k);
  m = cosd(angle);
  n = sind(angle);
  Qbar11 = Q11*m^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*n^4;
  Qbar22 = Q11*n^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*m^4;
  Qbar12 = (Q11 + Q22 - 4*Q66)*m^2*n^2 + Q12*(m^4 + n^4);
  Qbar16 = (Q11 - Q12 - 2*Q66)*m^3*n - (Q22 - Q12 - 2*Q66)*m*n^3;
  Qbar26 = (Q11 - Q12 - 2*Q66)*m*n^3 - (Q22 - Q12 - 2*Q66)*m^3*n;
  Qbar66 = (Q11 + Q22 - 2*Q12 - 2*Q66)*m^2*n^2 + Q66*(m^4 + n^4);
  Qbar = [Qbar11 Qbar12 Qbar16;
       Qbar12 Qbar22 Qbar26;
       Qbar16 Qbar26 Qbar66];
  A = A + Qbar * ply_thickness;
end
midplane_strain = A \ force_vector;
stress_results_A = zeros(number_of_plies,8);
for k = 1:number_of_plies
  angle = theta(k);
  m = cosd(angle);
  n = sind(angle);
  Qbar11 = Q11*m^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*n^4;
  Qbar22 = Q11*n^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*m^4;
  Qbar12 = (Q11 + Q22 - 4*Q66)*m^2*n^2 + Q12*(m^4 + n^4);
  Qbar16 = (Q11 - Q12 - 2*Q66)*m^3*n - (Q22 - Q12 - 2*Q66)*m*n^3;
```

```
  Qbar26 = (Q11 - Q12 - 2*Q66)*m*n^3 - (Q22 - Q12 - 2*Q66)*m^3*n;
  Qbar66 = (Q11 + Q22 - 2*Q12 - 2*Q66)*m^2*n^2 + Q66*(m^4 + n^4);
  Qbar = [Qbar11 Qbar12 Qbar16;
        Qbar12 Qbar22 Qbar26;
        Qbar16 Qbar26 Qbar66];
  stress_global = Qbar * midplane_strain;
  strain_x = midplane_strain(1);
  strain_y = midplane_strain(2);
  gamma_xy = midplane_strain(3);
  strain_1 = m^2*strain_x + n^2*strain_y + m*n*gamma_xy;
  strain_2 = n^2*strain_x + m^2*strain_y - m*n*gamma_xy;
  gamma_12 = -2*m*n*strain_x + 2*m*n*strain_y + (m^2 - n^2)*gamma_xy;
  stress_local = Q * [strain_1; strain_2; gamma_12];
  stress_results_A(k,:) = [k angle stress_global' stress_local'];
end
table_A = array2table(stress_results_A, ...
  'VariableNames', {'Ply','Angle_deg','sigma_x','sigma_y','tau_xy','sigma_1','sigma_2','tau_12'});
fprintf('\nProblem 2 Part A Results\n')
disp(table_A)
writetable(table_A, 'Problem2_PartA.xlsx');
%% Problem 2 Part B
% Layup: [+XX/-55/-55/+XX]
theta = [XX -55 -55 XX];
number_of_plies = length(theta);
A = zeros(3,3);
for k = 1:number_of_plies
  angle = theta(k);
  m = cosd(angle);
  n = sind(angle);
  Qbar11 = Q11*m^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*n^4;
  Qbar22 = Q11*n^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*m^4;
  Qbar12 = (Q11 + Q22 - 4*Q66)*m^2*n^2 + Q12*(m^4 + n^4);
  Qbar16 = (Q11 - Q12 - 2*Q66)*m^3*n - (Q22 - Q12 - 2*Q66)*m*n^3;
  Qbar26 = (Q11 - Q12 - 2*Q66)*m*n^3 - (Q22 - Q12 - 2*Q66)*m^3*n;
  Qbar66 = (Q11 + Q22 - 2*Q12 - 2*Q66)*m^2*n^2 + Q66*(m^4 + n^4);
  Qbar = [Qbar11 Qbar12 Qbar16;
        Qbar12 Qbar22 Qbar26;
        Qbar16 Qbar26 Qbar66];
  A = A + Qbar * ply_thickness;
end
midplane_strain = A \ force_vector;
stress_results_B = zeros(number_of_plies,8);
for k = 1:number_of_plies
  angle = theta(k);
  m = cosd(angle);
  n = sind(angle);
  Qbar11 = Q11*m^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*n^4;
  Qbar22 = Q11*n^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*m^4;
  Qbar12 = (Q11 + Q22 - 4*Q66)*m^2*n^2 + Q12*(m^4 + n^4);
  Qbar16 = (Q11 - Q12 - 2*Q66)*m^3*n - (Q22 - Q12 - 2*Q66)*m*n^3;
  Qbar26 = (Q11 - Q12 - 2*Q66)*m*n^3 - (Q22 - Q12 - 2*Q66)*m^3*n;
  Qbar66 = (Q11 + Q22 - 2*Q12 - 2*Q66)*m^2*n^2 + Q66*(m^4 + n^4);
  Qbar = [Qbar11 Qbar12 Qbar16;
        Qbar12 Qbar22 Qbar26;
        Qbar16 Qbar26 Qbar66];
  stress_global = Qbar * midplane_strain;
  strain_x = midplane_strain(1);
  strain_y = midplane_strain(2);
```

```
  gamma_xy = midplane_strain(3);
  strain_1 = m^2*strain_x + n^2*strain_y + m*n*gamma_xy;
  strain_2 = n^2*strain_x + m^2*strain_y - m*n*gamma_xy;
  gamma_12 = -2*m*n*strain_x + 2*m*n*strain_y + (m^2 - n^2)*gamma_xy;
  stress_local = Q * [strain_1; strain_2; gamma_12];
  stress_results_B(k,:) = [k angle stress_global' stress_local'];
end
table_B = array2table(stress_results_B, ...
  'VariableNames', {'Ply','Angle_deg','sigma_x','sigma_y','tau_xy','sigma_1','sigma_2','tau_12'});
fprintf('\nProblem 2 Part B Results\n')
disp(table_B)
writetable(table_B, 'Problem2_PartB.xlsx');
%% Problem 2 Part C and D
% Find stresses in each ply for each alpha
% Then plot sigma_x in each ply as a function of alpha
alpha_values = 3:3:30;
all_partC_results = [];
sigma_x_ply1 = zeros(length(alpha_values),1);
sigma_x_ply2 = zeros(length(alpha_values),1);
sigma_x_ply3 = zeros(length(alpha_values),1);
sigma_x_ply4 = zeros(length(alpha_values),1);
for i = 1:length(alpha_values)
  alpha = alpha_values(i);
  theta = [XX -alpha -alpha XX];
  number_of_plies = length(theta);
  A = zeros(3,3);
  for k = 1:number_of_plies
    angle = theta(k);
    m = cosd(angle);
    n = sind(angle);
    Qbar11 = Q11*m^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*n^4;
    Qbar22 = Q11*n^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*m^4;
    Qbar12 = (Q11 + Q22 - 4*Q66)*m^2*n^2 + Q12*(m^4 + n^4);
    Qbar16 = (Q11 - Q12 - 2*Q66)*m^3*n - (Q22 - Q12 - 2*Q66)*m*n^3;
    Qbar26 = (Q11 - Q12 - 2*Q66)*m*n^3 - (Q22 - Q12 - 2*Q66)*m^3*n;
    Qbar66 = (Q11 + Q22 - 2*Q12 - 2*Q66)*m^2*n^2 + Q66*(m^4 + n^4);
    Qbar = [Qbar11 Qbar12 Qbar16;
            Qbar12 Qbar22 Qbar26;
            Qbar16 Qbar26 Qbar66];
    A = A + Qbar * ply_thickness;
  end
  midplane_strain = A \ force_vector;
  for k = 1:number_of_plies
    angle = theta(k);
    m = cosd(angle);
    n = sind(angle);
    Qbar11 = Q11*m^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*n^4;
    Qbar22 = Q11*n^4 + 2*(Q12 + 2*Q66)*m^2*n^2 + Q22*m^4;
    Qbar12 = (Q11 + Q22 - 4*Q66)*m^2*n^2 + Q12*(m^4 + n^4);
    Qbar16 = (Q11 - Q12 - 2*Q66)*m^3*n - (Q22 - Q12 - 2*Q66)*m*n^3;
    Qbar26 = (Q11 - Q12 - 2*Q66)*m*n^3 - (Q22 - Q12 - 2*Q66)*m^3*n;
    Qbar66 = (Q11 + Q22 - 2*Q12 - 2*Q66)*m^2*n^2 + Q66*(m^4 + n^4);
    Qbar = [Qbar11 Qbar12 Qbar16;
            Qbar12 Qbar22 Qbar26;
            Qbar16 Qbar26 Qbar66];
    stress_global = Qbar * midplane_strain;
    sigma_x = stress_global(1);
    sigma_y = stress_global(2);
```

```
        tau_xy = stress_global(3);
        strain_x = midplane_strain(1);
        strain_y = midplane_strain(2);
        gamma_xy = midplane_strain(3);
        strain_1 = m^2*strain_x + n^2*strain_y + m*n*gamma_xy;
        strain_2 = n^2*strain_x + m^2*strain_y - m*n*gamma_xy;
        gamma_12 = -2*m*n*strain_x + 2*m*n*strain_y + (m^2 - n^2)*gamma_xy;
        stress_local = Q * [strain_1; strain_2; gamma_12];
        sigma_1 = stress_local(1);
        sigma_2 = stress_local(2);
        tau_12 = stress_local(3);
        all_partC_results = [all_partC_results;
          alpha k angle sigma_x sigma_y tau_xy sigma_1 sigma_2 tau_12];
        if k == 1
          sigma_x_ply1(i) = sigma_x;
        elseif k == 2
          sigma_x_ply2(i) = sigma_x;
        elseif k == 3
          sigma_x_ply3(i) = sigma_x;
        elseif k == 4
          sigma_x_ply4(i) = sigma_x;
        end
    end
end
table_C = array2table(all_partC_results, ...
  'VariableNames', {'alpha_deg','Ply','Angle_deg','sigma_x','sigma_y','tau_xy','sigma_1','sigma_2','tau_12'});
fprintf('\nProblem 2 Part C Results\n')
disp(table_C)
writetable(table_C, 'Problem2_PartC.xlsx');
%% Problem 2 Part D: Final Plot
figure
plot(alpha_values, sigma_x_ply1, 'o-', 'LineWidth', 1.5)
hold on
plot(alpha_values, sigma_x_ply2, 's-', 'LineWidth', 1.5)
plot(alpha_values, sigma_x_ply3, 'd-', 'LineWidth', 1.5)
plot(alpha_values, sigma_x_ply4, '^-', 'LineWidth', 1.5)
grid on
xlabel('\alpha [deg]')
ylabel('\sigma_x [MPa]')
title('Problem 2: \sigma_x in Each Ply vs \alpha')
legend('Ply 1','Ply 2','Ply 3','Ply 4','Location','best')
sigma_x_all = table_C.sigma_x;
[largest_sigma_x, index] = max(sigma_x_all);
best_alpha = table_C.alpha_deg(index);
best_ply = table_C.Ply(index);
plot(best_alpha, largest_sigma_x, 'ko', ...
  'MarkerSize', 10, 'LineWidth', 2)
text(best_alpha, largest_sigma_x, ...
  sprintf('  Max: \\alpha = %d^\\circ, Ply %d', best_alpha, best_ply))
fprintf('\nMaximum sigma_x occurs at alpha = %d degrees\n', best_alpha)
fprintf('Maximum sigma_x occurs in ply %d\n', best_ply)
fprintf('Maximum sigma_x = %.3f MPa\n', largest_sigma_x)
```